VICTORY 2027

Sense Organs

LECTURE-2 BIOLOGY

Upasana ma'am



Topics To Be Covered

1. Structure and Working of Eye.
2.


HELLO
NINJAS
忍者参上
影の中で戦う
忍者
忍術最強
忍者
忍者参上

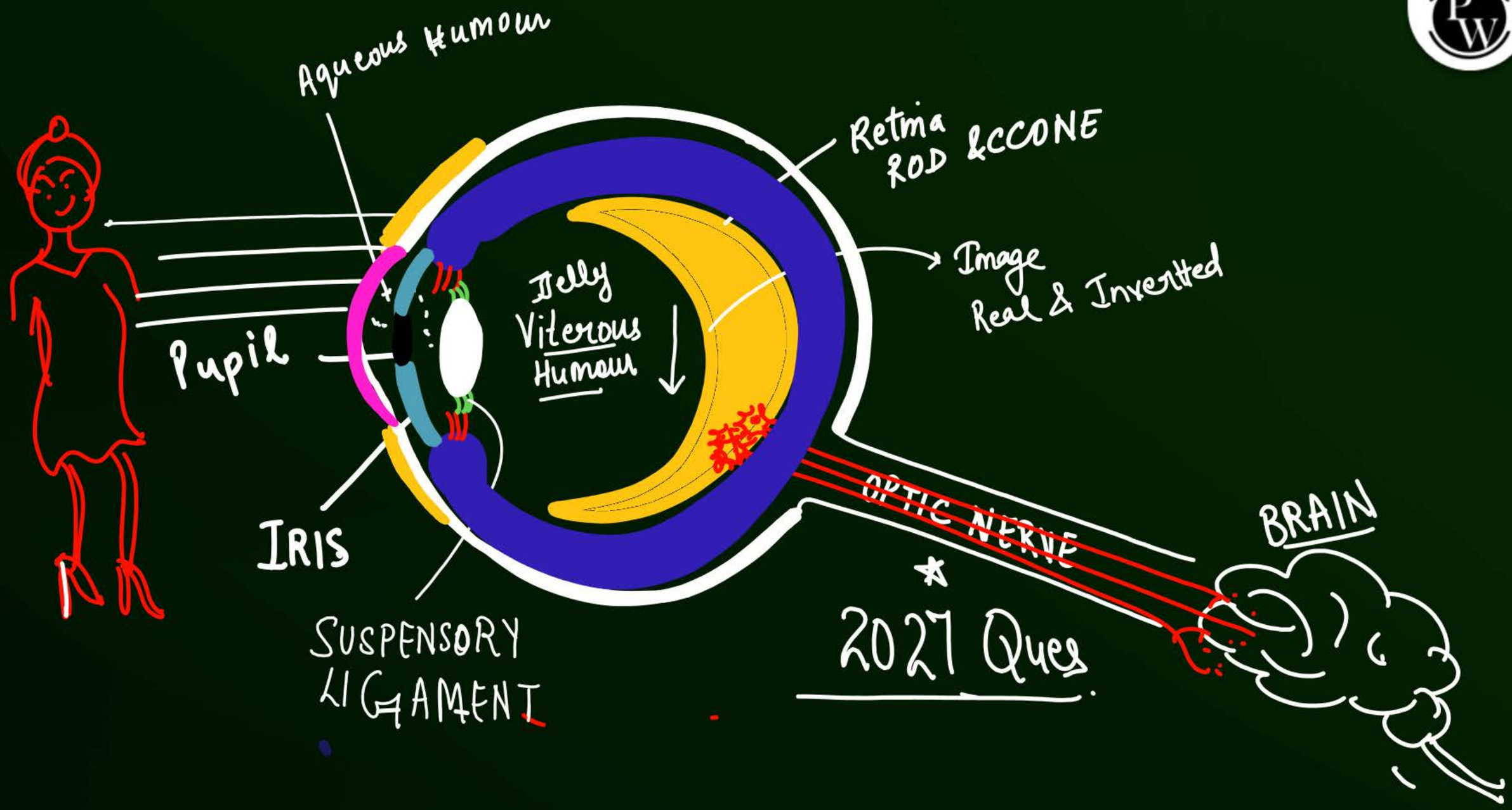
RATTON KO KARO **BYE BYE !!**

EASY CONCEPTS SE KARO **HELLO HI !!**

With – *Uppu Ma'am*




Sclera
Cornea
Pupil
Iris
Lens
Ciliary muscles
Vitreous
Retina
(cones &rods)
Optic Nerve

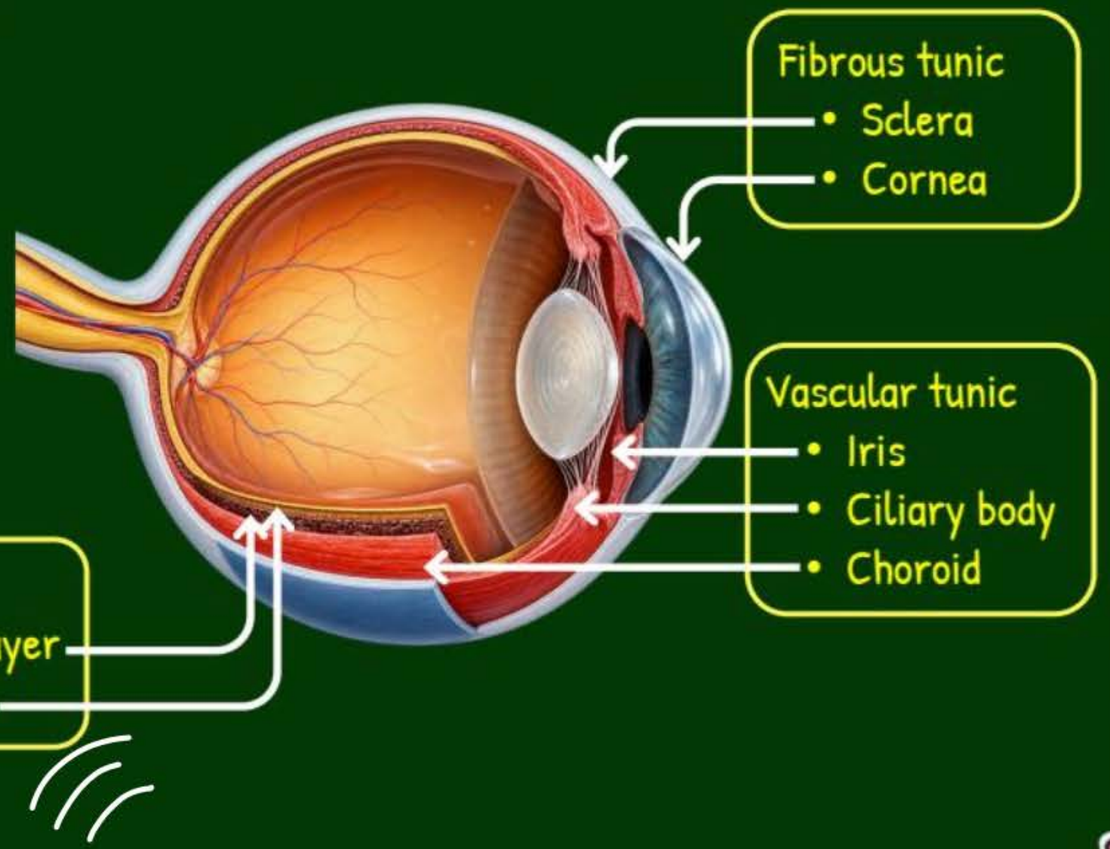

Aqueous Humour
Retina
ROD & CONE
Image
Real & Inverted
Pupil
Jelly
Viterous
Humour
IRIS
OPTIC NERVE
BRAIN
SUSPENSORY
LIGAMENT
2027 Ques.

STRUCTURE OF THE EYEBALL

- The wall of the eyeball is composed of three concentric layers :

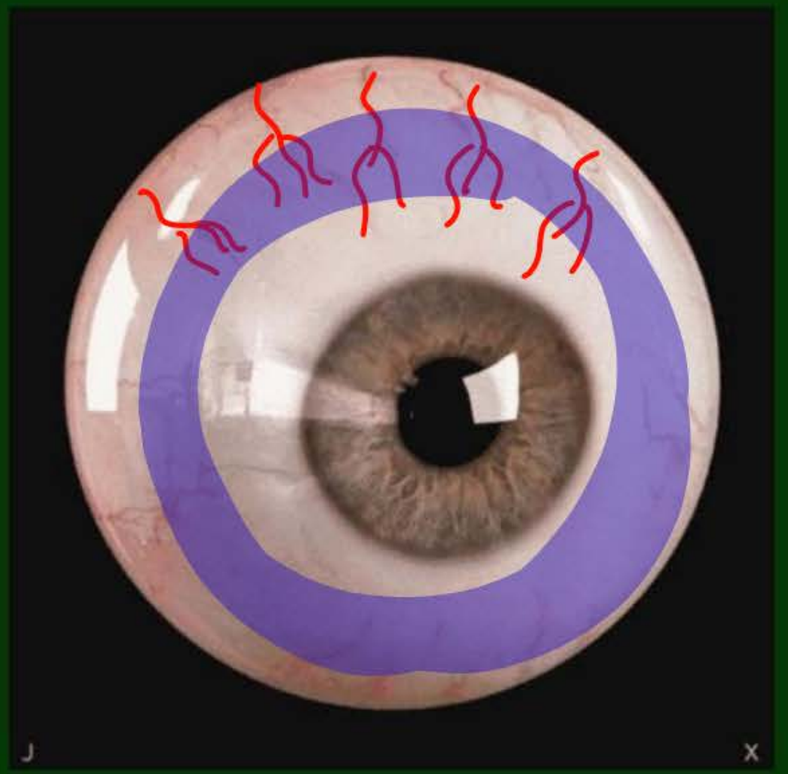
1. Outer sclerotic,
2. Middle choroid, and
3. Inner retina.

Anatomical structure



STRUCTURE OF THE EYEBALL

- The sclerotic layer (or sclera) is made of tough fibrous tissues and is white in colour.
- Visible through the conjunctiva.
- It bulges out and becomes transparent in the front region where it covers the coloured part of the eye called cornea.



DONATION OF EYE

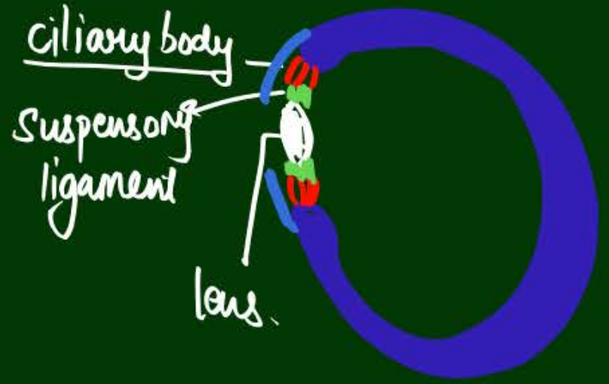
- Cornea remains alive up to nearly 40 hours after the death of a person. If donated, the eye can be removed soon (within about 4 hours) after death. It is stored in an eye bank at a very low temperature in a suitable medium such as blood plasma or certain other sterile (germ-free) liquid medium. For grafting, the cornea part is taken out from the eye and is fixed in place of the defective cornea and the vision is restored.

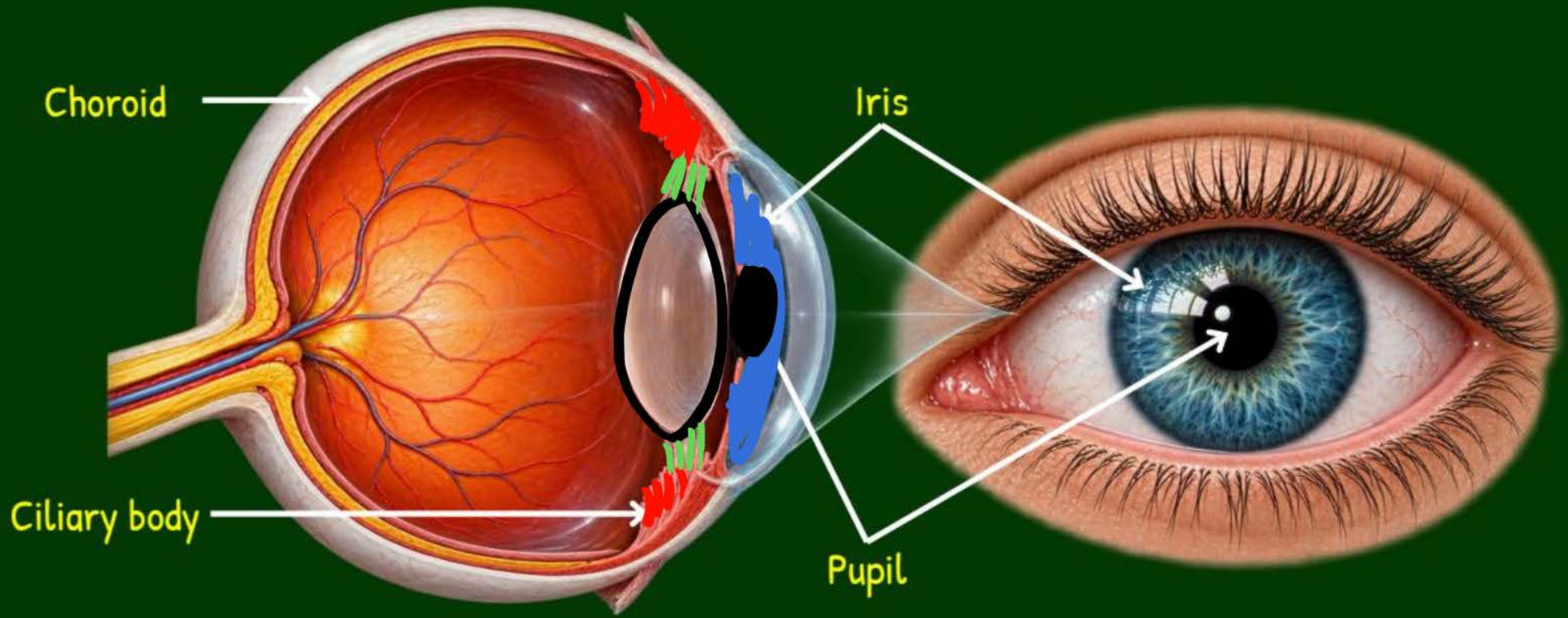


'CHOROID'

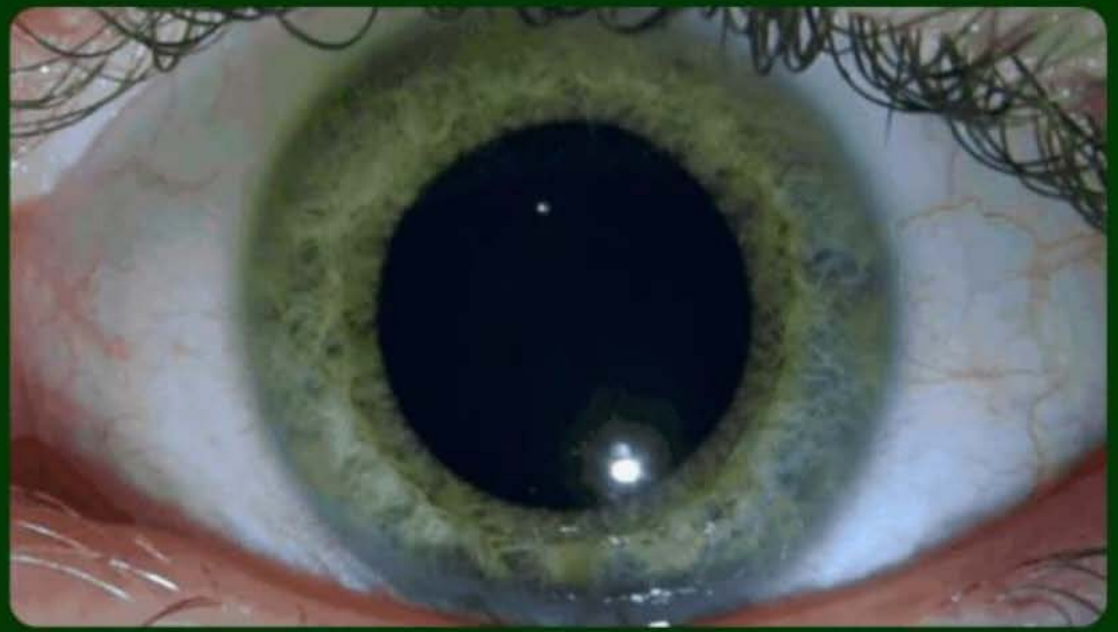
- The choroid layer is richly supplied with blood vessels for providing nourishment to the eye.
- It contains a dark black pigment (melanin) which prevents light rays from reflecting and scattering inside the eye.
- In the front of the eye, the choroid expands to form the ciliary body (containing circular muscles).
- The (smooth muscles) in the ciliary body alter the shape of the lens.

['involuntary' muscles]

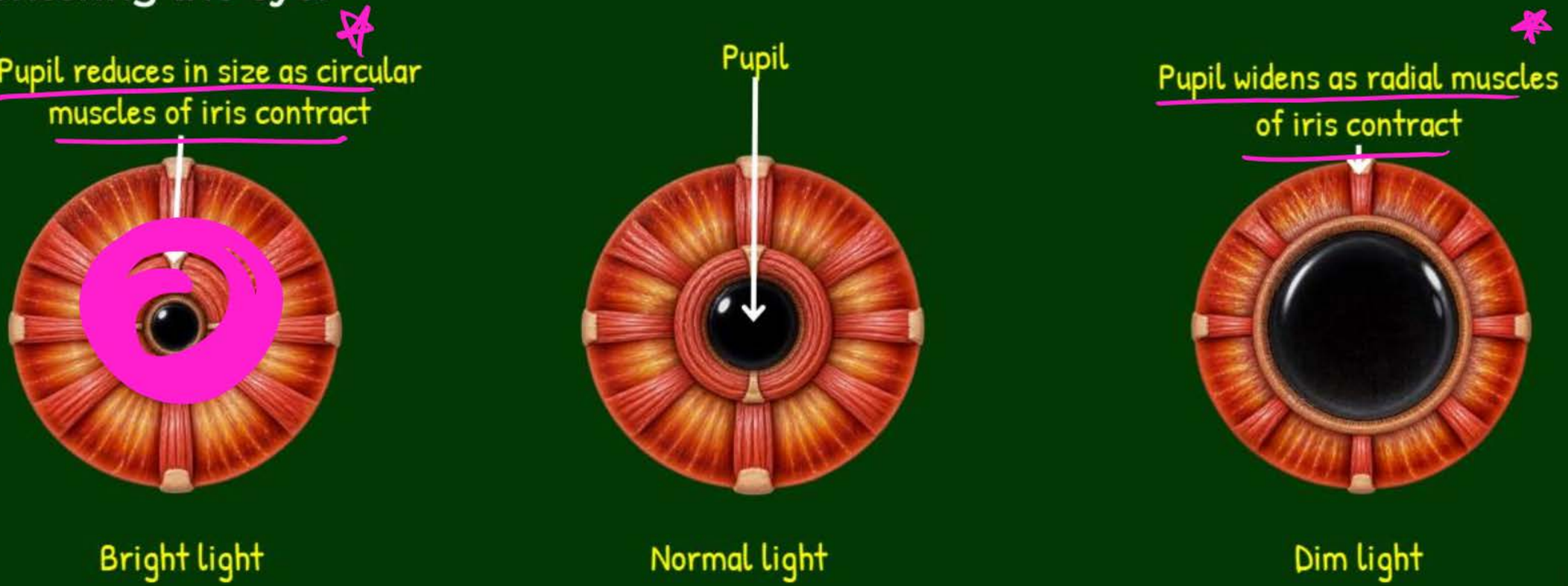


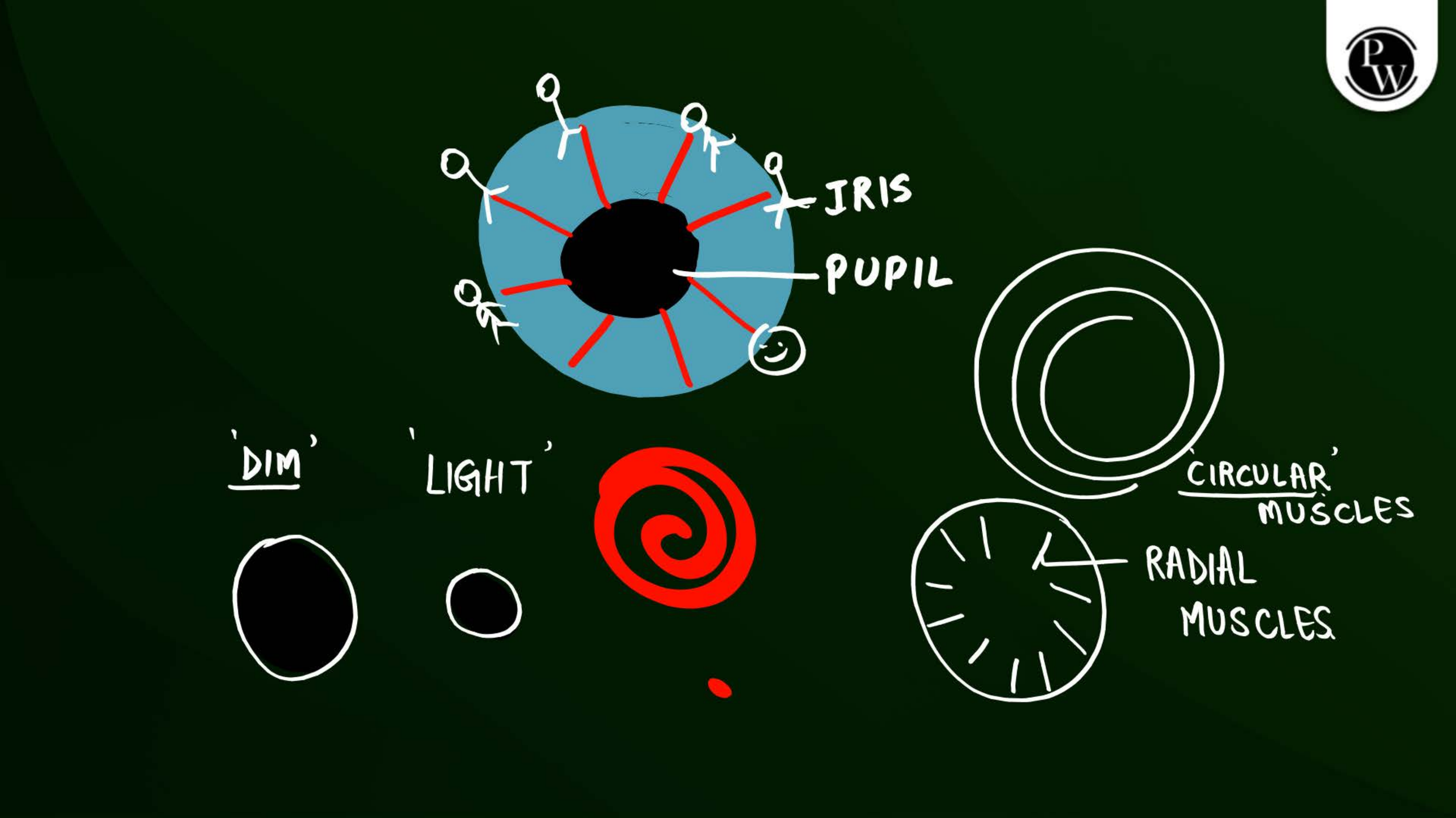

Choroid
Iris
Ciliary body
Pupil

- The iris (Latin iris : rainbow) is also an extension of the choroid,
- It covers the lens partially
- It has a circular opening in the centre called pupil.
- The blue, brown or black colour of the eye refers to the colour of the iris.
- ["PUPIL" name has been derived from the Latin word "pupa" meaning "doll",



- The iris contains radial muscles to widen and circular muscles to constrict the pupil. This adjustment of the size of the pupil regulates the amount of light entering the eye.




IRIS
PUPIL
'DIM'
'LIGHT'
'CIRCULAR' MUSCLES
RADIAL MUSCLES

QUESTIONS

The layer in the eye where sensory cells (rods and cones) are located:

A Conjunctiva

B Cornea

C Choroid

D Retina ✓

QUESTIONS

An aperture that controls the passage of light into the eye is:

A. Blind spot

B. Pupil

C. Yellow spot

D. Iris

QUESTIONS

Tears have an antiseptic property due to the presence of:

A Lysosome

B Aqueous humour

C Lysozyme

D Vitreous humour

QUESTIONS

Which of the following is responsible for the adjustment of the size of pupil?

A Iris

B Sclera

C Lens

D Choroid

QUESTIONS

The thin, transparent extension of sclerotic layer found in front of the lens is:

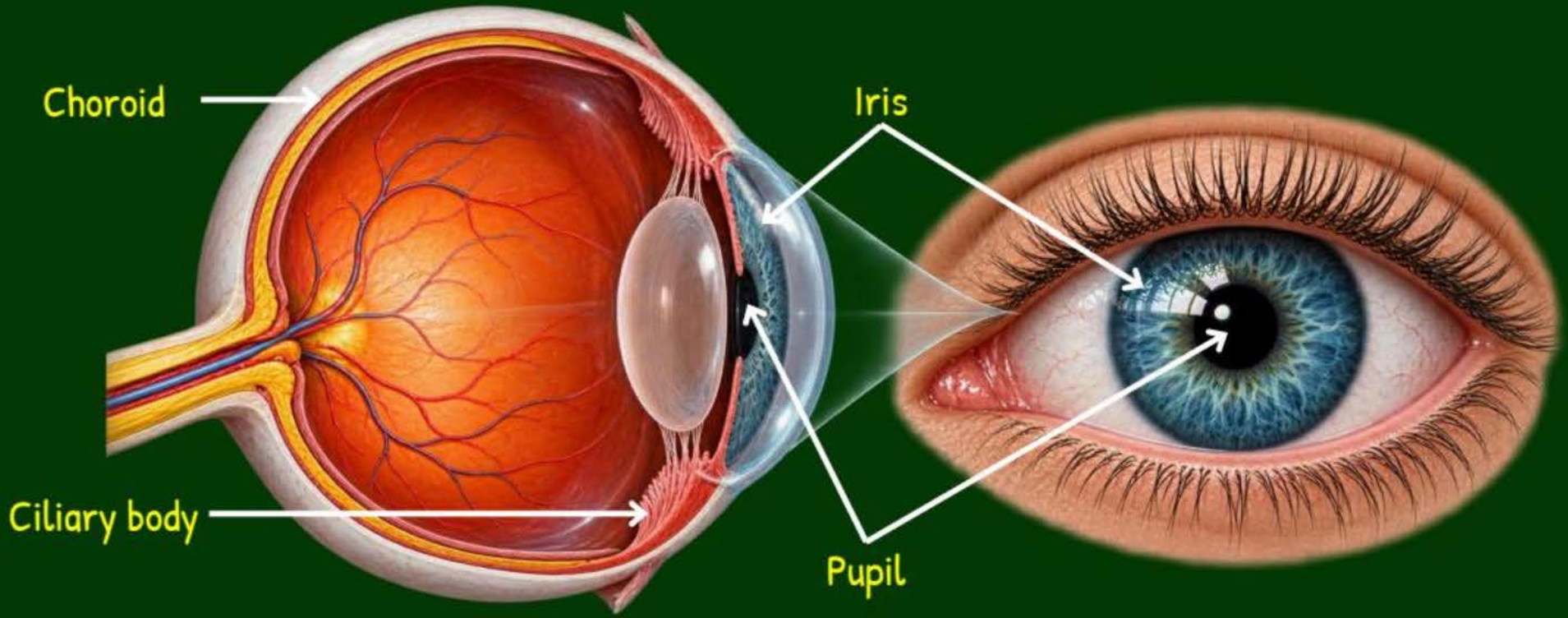
A Cornea

B Cochlea

C Conjunctiva

D Choroid

- The retina or the innermost layer is sensitive to light.
- It contains two types of sense cells called rods and cones.



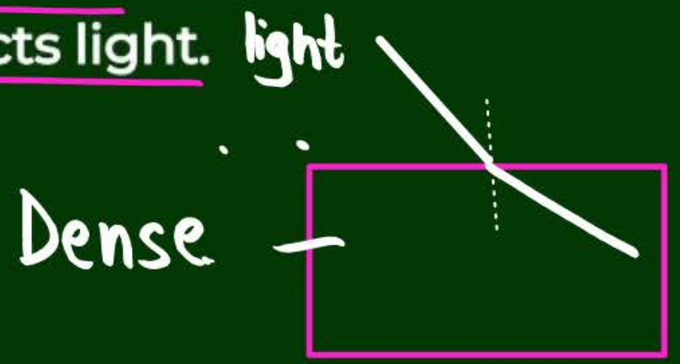
- Two chambers of the eye aqueous and vitreous chambers

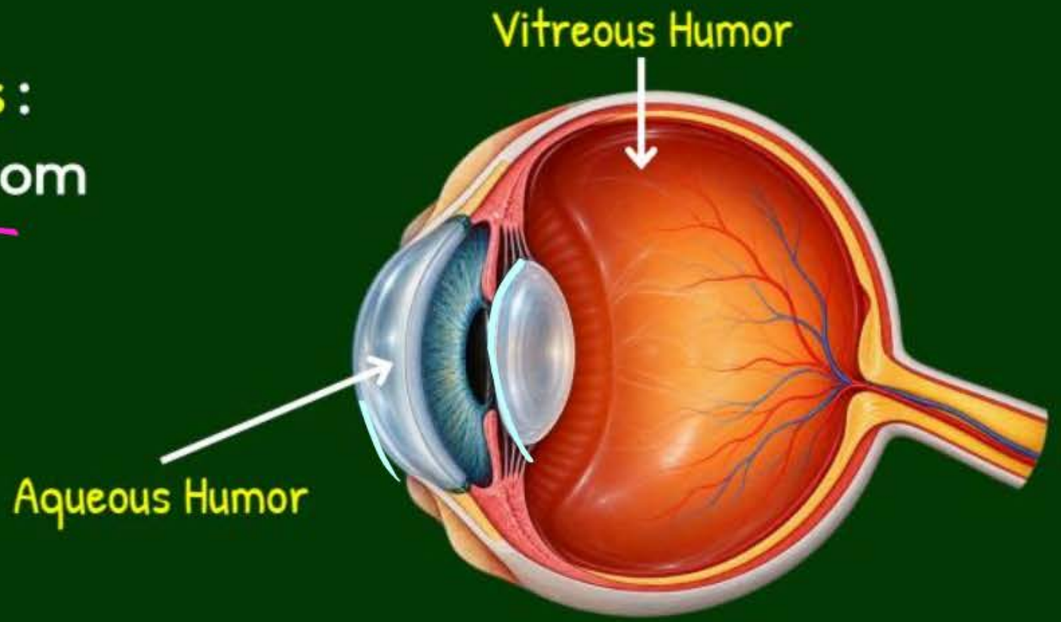
1. Aqueous chamber is the front chamber between the lens and the cornea. It is filled with a clear watery liquid called aqueous humour (aqueous : watery; humour : fluid).

The aqueous humour serves in two ways :

(i) Keeps the lens moist and protects it from physical shock,

(ii) It refracts light.



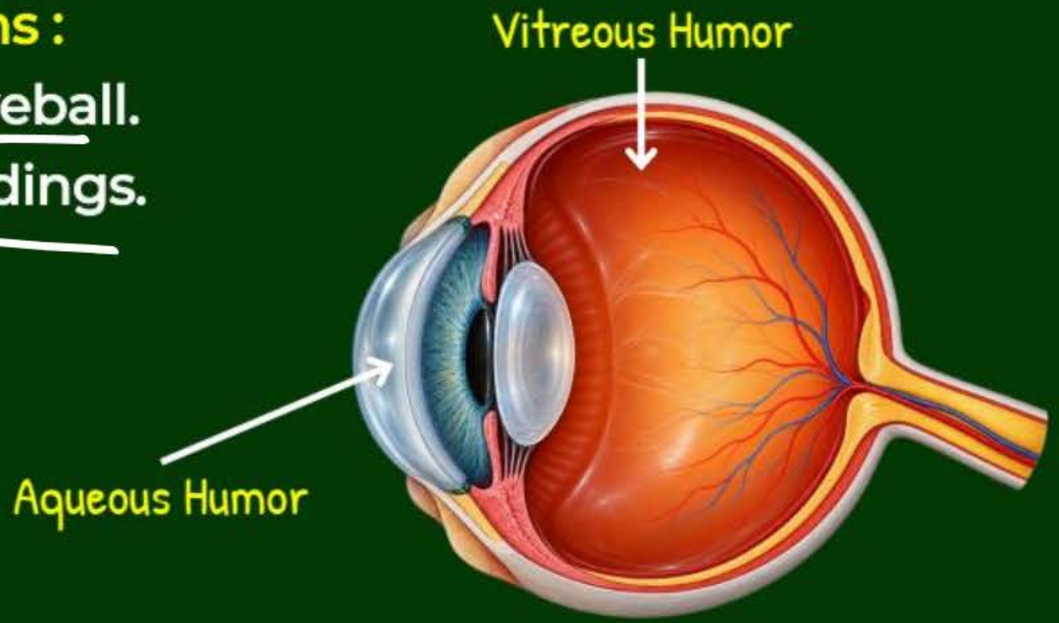


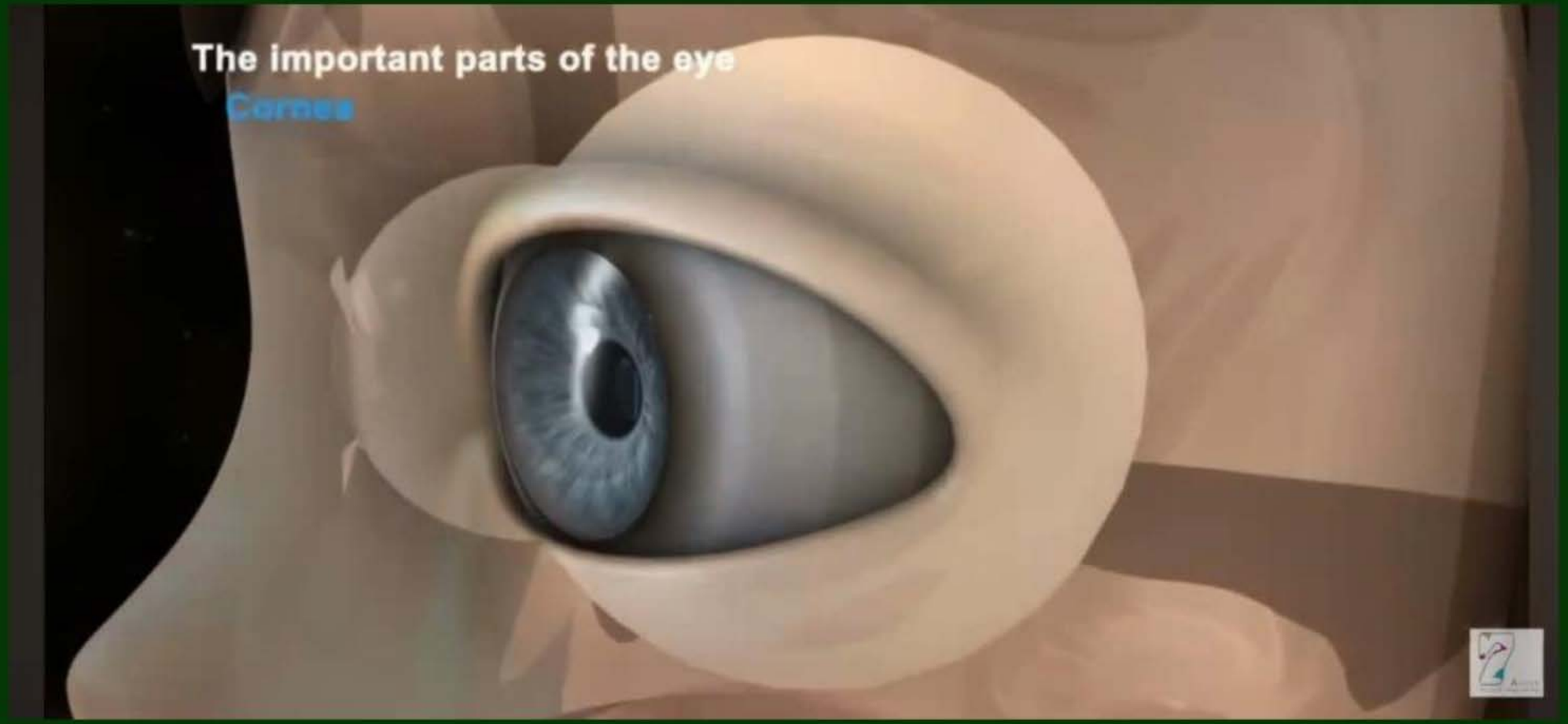
1. Vitreous chamber is the larger cavity of the eyeball behind the lens. It is filled with transparent jelly-like thicker fluid called vitreous humour (vitreous : glassy; humour : fluid).

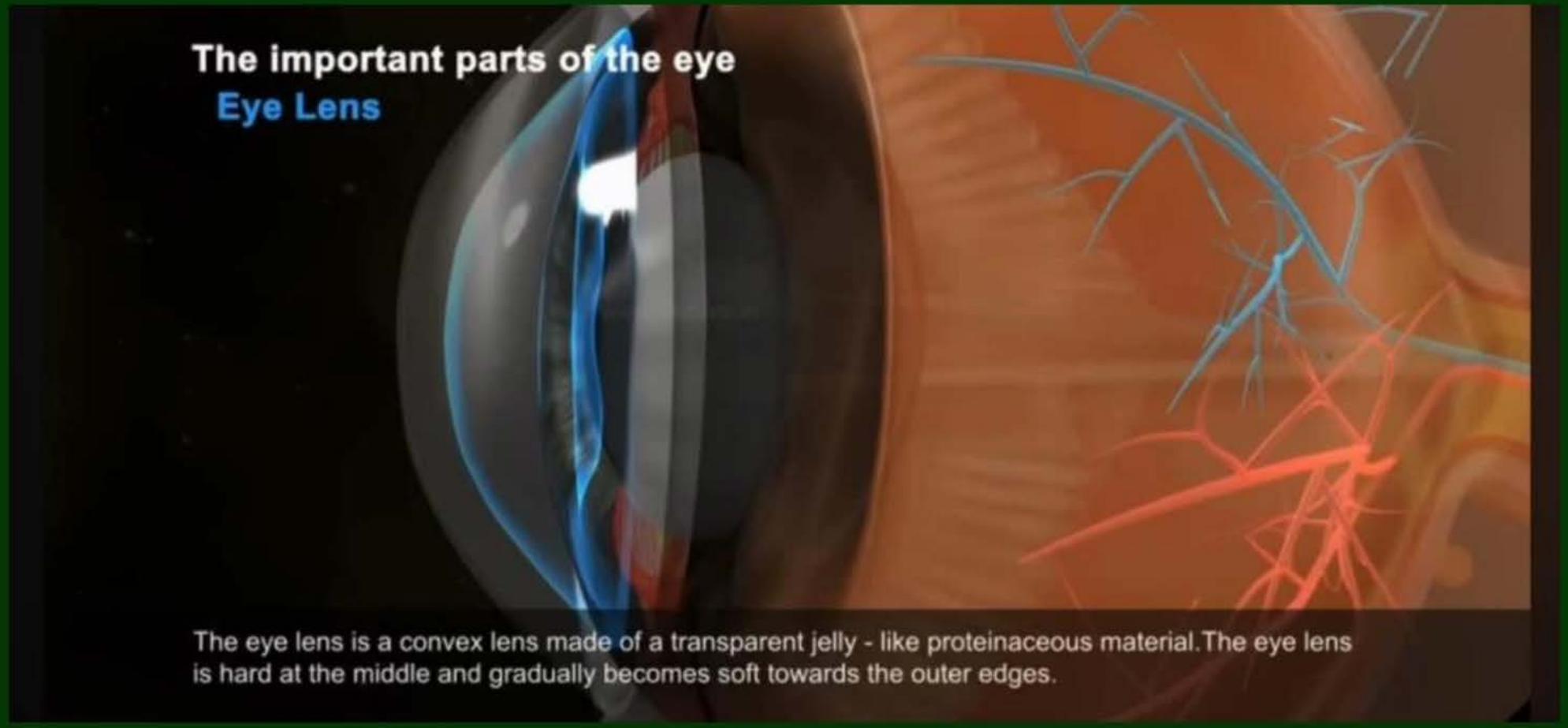
The vitreous humour serves two functions :

(i) It helps in keeping the shape of the eyeball.

(ii) It protects the retina and its nerve endings.




The important parts of the eye
Cornea


The important parts of the eye
Eye Lens
The eye lens is a convex lens made of a transparent jelly - like proteinaceous material.The eye lens is hard at the middle and gradually becomes soft towards the outer edges.

HOW DO WE SEE?

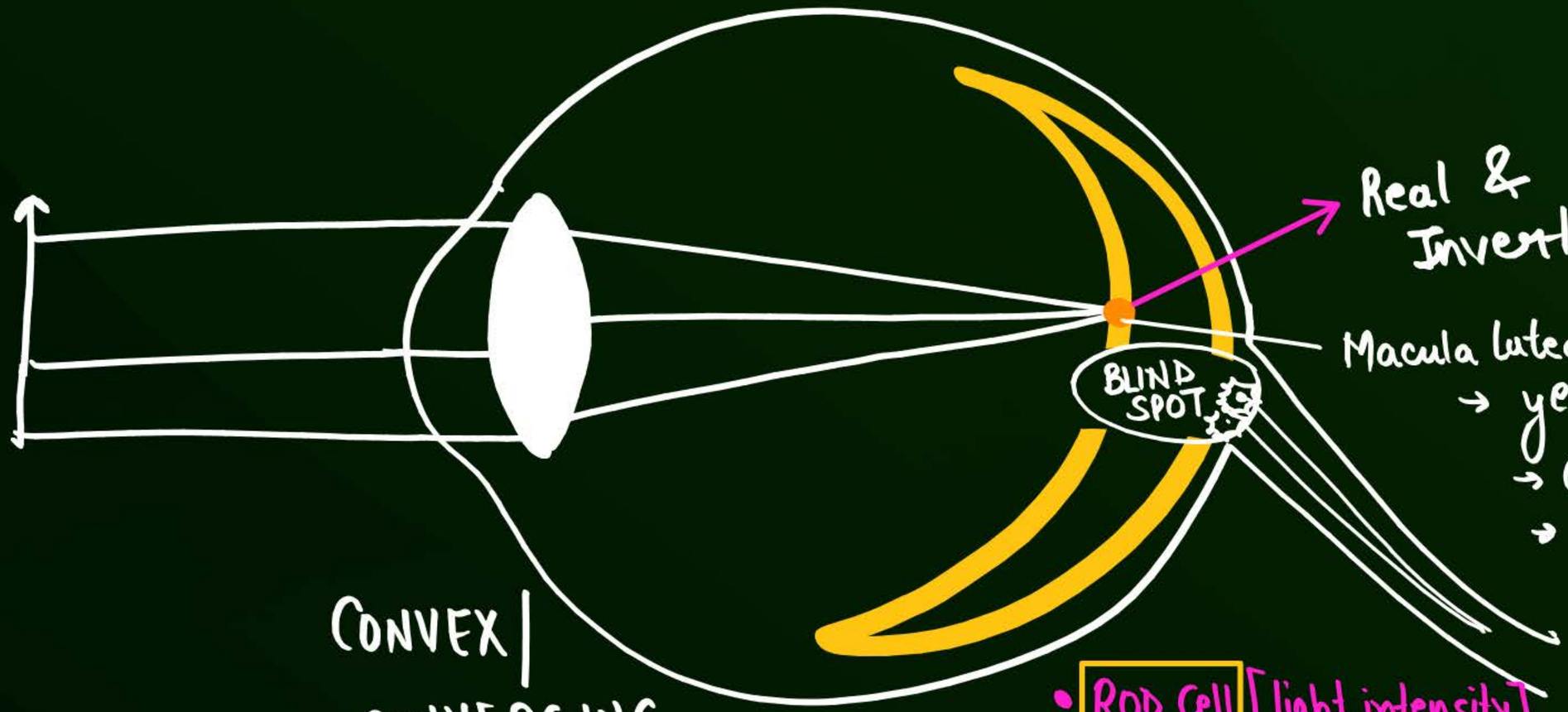
The four major steps in seeing an object are as follows:

(1) **Entry of light rays**: Light rays from the object enter the eyes through the transparent structures (conjunctiva, cornea, aqueous humour, lens, vitreous humour).→ Retina ✓imp

(2) **Focusing of image** : First, the curvature of the cornea converges the light rays to some extent and the lens converges them further to form an image on the retina. The image on the retina is inverted and real.

(3) **Transmission of nerve impulses from retina to brain :** due to chemical changes in cones and rod cells , nerve impulse is sent to brain through optic nerve

(4) **Interpretation by the brain :** image formed in the eye is inverted.

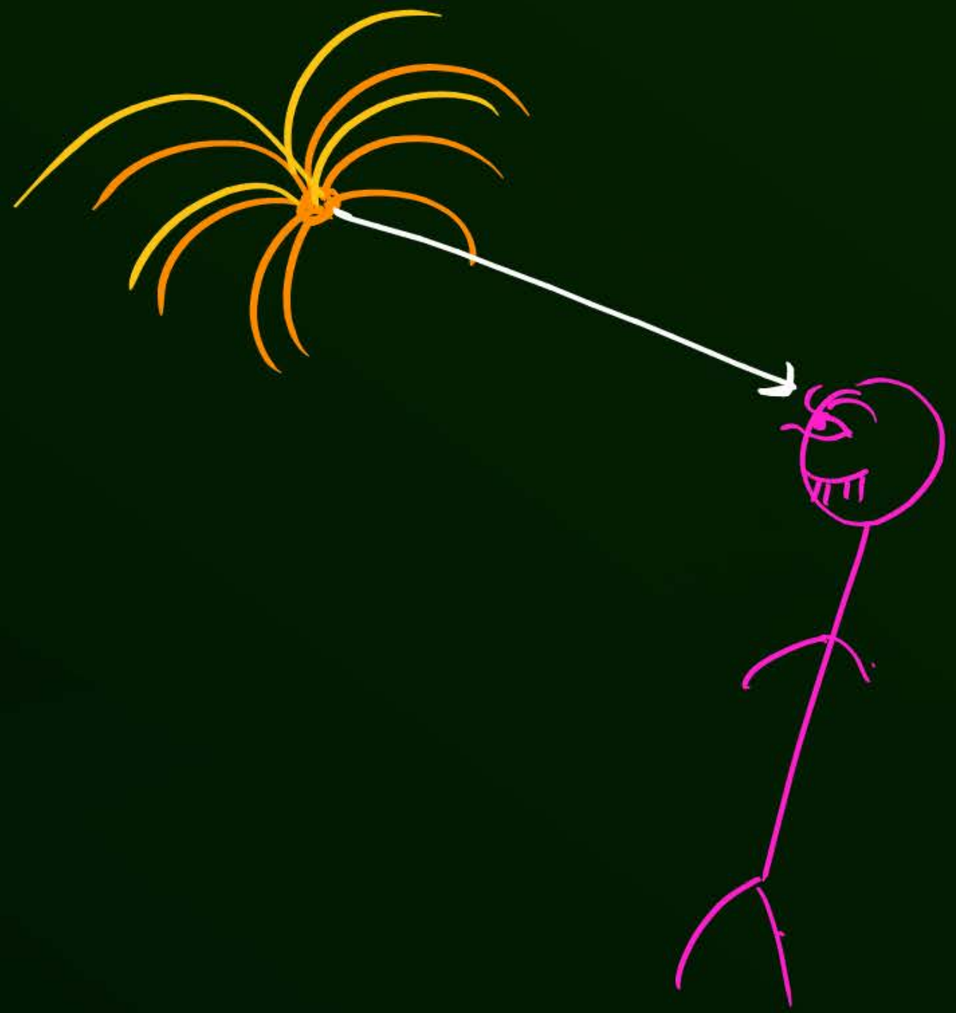


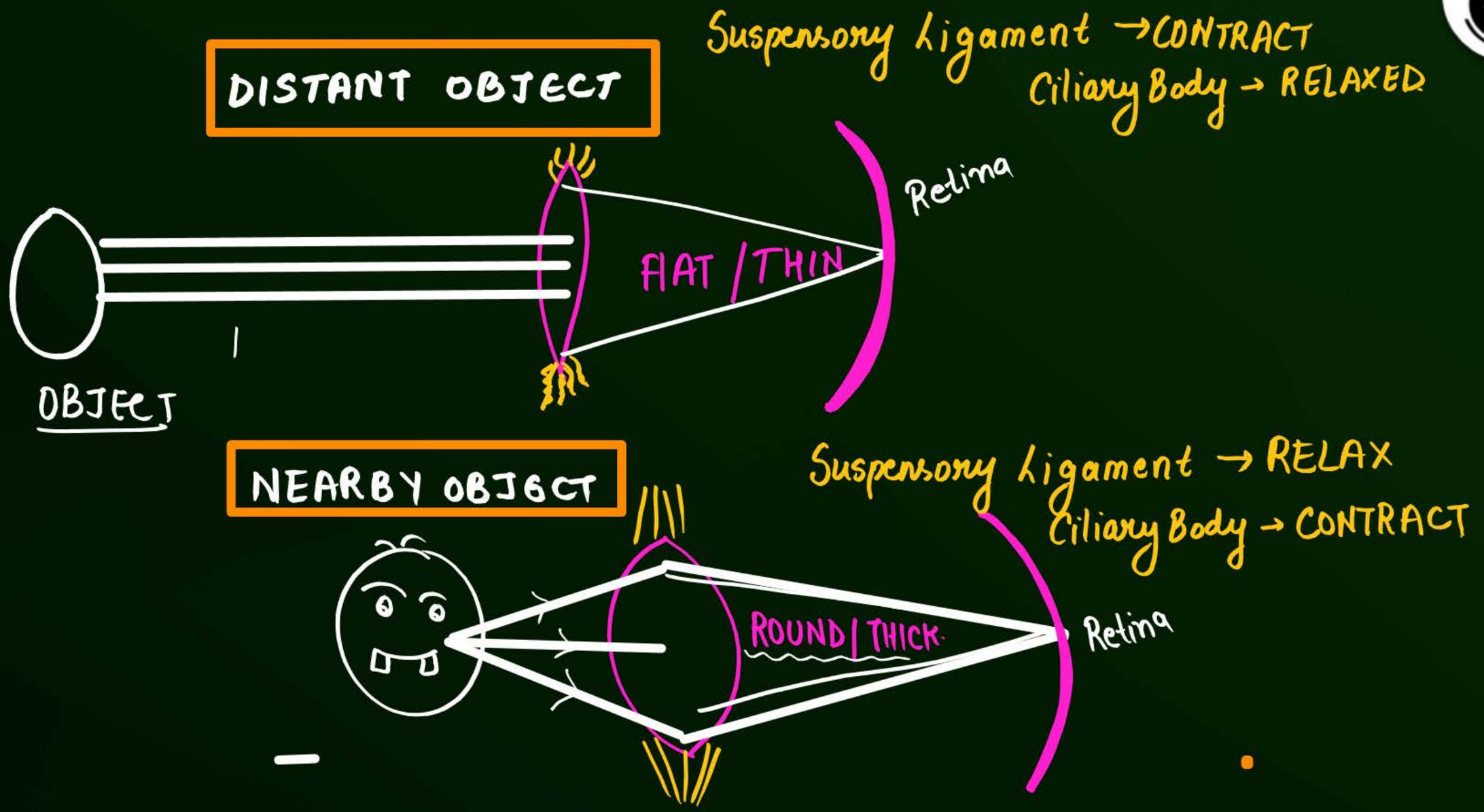
Real & Inverted

Macula lutea / fovea
→ yellow point
→ CONE Cells (more)
→ sharp vision

CONVEX / CONVERGING LENS

- ROD Cell [light intensity].
  More activated during dim light.
  20-times more than CONE CELLS.

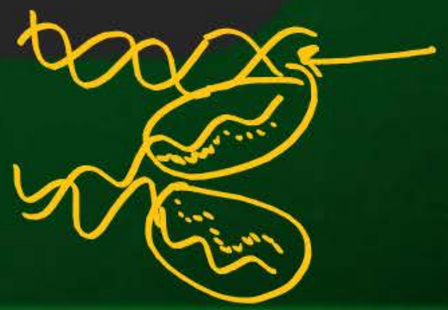



DISTANT OBJECT
Suspensory Ligament → CONTRACT
Ciliary Body → RELAXED
Retina
FLAT / THIN
OBJECT
NEARBY OBJECT
Suspensory Ligament → RELAX
Ciliary Body → CONTRACT
ROUND / THICK
Retina

Thank You







ONE SHOT

QAns

Answer learning

ciliary